Exercise 2

① SELECT DISTINCT department
FROM students;

| department |
|---|
| IT |
| HR |
| Finance |

② SELECT department,
AVG (age) AS avg_age
FROM Students
GROUP BY department;

| department | avg_age |
|---|---|
| IT | 20,5 |
| HR | 22 |
| Finance | 25 |

③ SELECT department,
count (*) AS student_count
FROM Students
GROUP BY department
HAVING COUNT(*) >1;

| department | student_count |
|---|---|
| IT | 2 |
| HR | 2 |

④ SELECT student_id,
name,
age,
department
FROM students
WHERE age BETWEEN 21 AND 23;

| student_id | name | age | department |
|---|---|---|---|
| 2 | Bob | 22 | HR |
| 3 | Charlie | 21 | IT |
| 4 | Diana | 23 | Finance |
| 5 | Eve | 22 | HR |

(5)
```
SELECT student_id,
       name,
       age,
       department
FROM students
WHERE department IN ('IT','HR') AND age>=21
```

| student_id | name | age | department |
|---|---|---|---|
| 2 | Bob | 22 | HR |
| 5 | Eve | 22 | HR |

(6)
```
SELECT department,
       SUM(credits) AS total_credits
FROM students
GROUP BY department
HAVING SUM(credits) >5
```

| department | total_credits |
|---|---|
| IT | 11 |

(7)
```
SELECT course_id,
       course_name,
       department,
       credits
FROM courses
WHERE credits < 4
```

| course_id | course_name | department | credits |
|---|---|---|---|
| 101 | SQL Basics | IT | 3 |
| 104 | Excel | Finance | 2 |
| 105 | Statistics | HR | 3 |

(8)
```
SELECT course_id,
       course_name,
       credits
FROM courses
ORDER BY Credits DESC
Limit 3;
```

| course_id | course_name | credits |
|---|---|---|
| 102 | Python | 4 |
| 103 | Data Science | 4 |
| 101 | SQL Basics | 3 |

(9)
```
SELECT max(grade) AS max_grade,
       min(grade) AS min_grade,
       avg(grade) AS avg_grade
FROM enrollments;
```

| max_grade | min_grade | avg_grade |
|---|---|---|
| 90 | 78 | 84,6 |

10
```
SELECT course_id,
       COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

| course_id | enrollment_count |
|---|---|
| 101 | 1 |
| 102 | 1 |
| 103 | 1 |
| 104 | 1 |
| 105 | 1 |

(11) select department,
Sum (salary) AS total_salary,
SUM (bonus) AS total_bonus
FROM Salaries
GROUP BY department;

| department | total_salary | total_bonus |
|---|---|---|
| IT | 122 000 | 10500 |
| HR | 101 000 | 7500 |
| Finance | 70000 | 6 000 |

(12) Select department,
AVG (salary) AS avg_salary
FROM Salaries
GROUP BY department
HAVING AVG (salary) > 56000;

| department | avg_salary |
|---|---|
| IT | 61 000 |
| Finance | 70 000 |

(13) Select employee_id,
name,
salary,
bonus,
salary + bonus AS total_compensation
FROM salaries
WHERE salary + bonus > 60000

| employee_id | name | salary | bonus |
|---|---|---|---|
| 1 | Tom | 60000 | 3000 |
| 3 | Spike | 70000 | 600 |
| 4 | Tyke | 62000 | 5500 |

(14) SELECT department,
SUM (budget) AS total_budget,
AVG (budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70000;

| department | total_budget | avg_budget |
|---|---|---|
| IT | 270000 | 135000 |
| Finance | 80000 | 80000 |

(15) Select project_id,
project_name,
department,
budget
FROM projects
WHERE budget BETWEEN 5000 AND 120000 AND department NOT
marketing;

| project_id | project_name | department | budget |
|---|---|---|---|
| 1 | AI App | IT | 100000 |
| 2 | Payroll system | Finance | 50000 |
| 5 | HR portal | HR | 50000 |